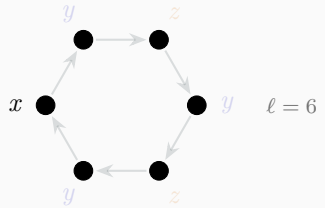
An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

Yuya Uezato Yoshiki Nakamura
CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

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FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



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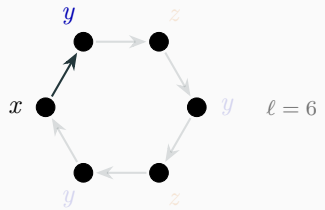
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- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
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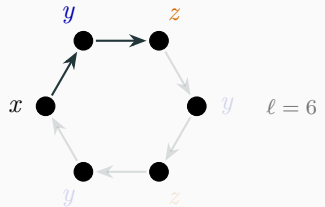
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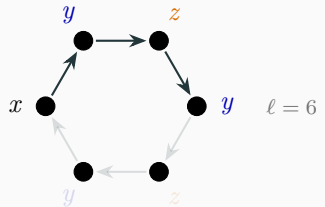
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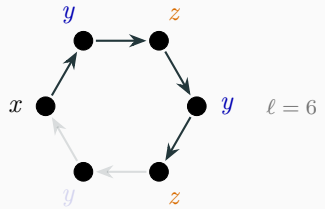
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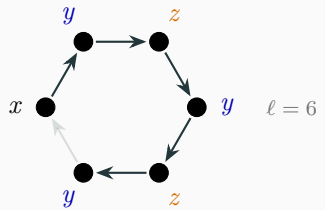
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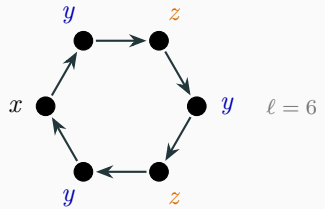
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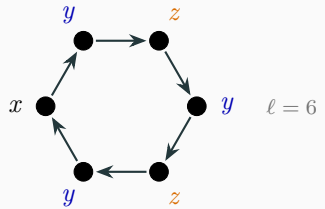
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We consider binary relation structures with n elements:

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FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
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Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

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To obtain the form of $\exists z.\ \cdots$, we use DNF translation.
$\rightsquigarrow 2^{O(\lvert\varphi\rvert)}$ compositions are used [NJ22, JLAMP].

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(This translation uses 2^k compositions, one for each subset $A \subseteq [k]$.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 *to* CoR*?*

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

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Outline

1. Our results

2. Technical overview

3. Conclusion

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There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

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- 答えは YES：$|\varphi_k| = O(k)$ で合成 $\geq 2^{k-1}$ 個
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More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
  exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
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- 答えは YES：$|\varphi_k| = O(k)$ で合成 $\geq 2^{k-1}$ 個
- 同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界

Our results

The answer is **YES!**

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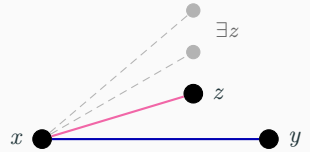
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The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



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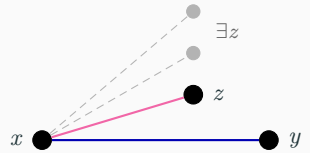
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Outline

1. Our results

2. Technical overview

3. Conclusion

Proof outline (high level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

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	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
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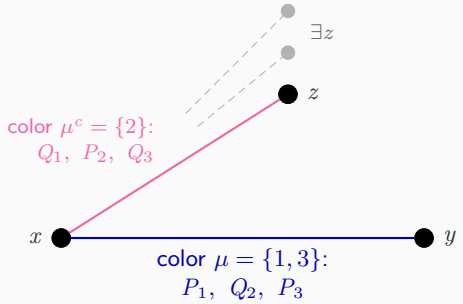
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Structures as colored complete graphs

View structures as **complete graphs**, colored with 2^k colors $\mu \subseteq [k]$: $P_i \iff i \in \mu$.

Q_i is used as the complement of P_i (i.e., $i \notin \mu$): exactly one of P_i, Q_i holds on each edge.



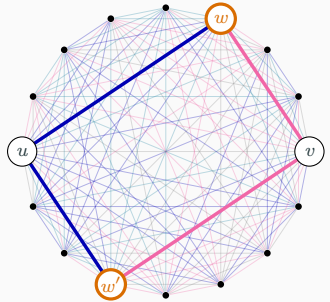
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☛ $G_k(x, y)$ holds iff some x-z edge has the color μ^c, where μ is the color of the x-y edge.

ページ概要：

- 構造を色付き完全グラフとして見る
- $G_k(x, y)$：x-y 辺の色の補の色の x-z 辺がある

The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k4^k)$ vertices,
cf. [JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** w such that the u-w and w-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

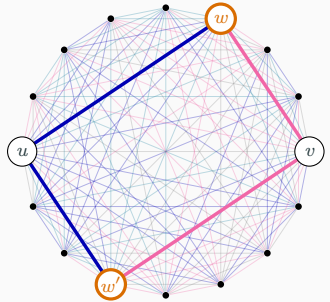
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Proof by induction on t: the case $t = t_1 \,;\, t_2$ follows from the (two-)witness property.

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- $\mathcal{M}_0$：ランダム彩色、two-witness property
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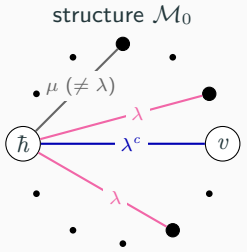
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Hub $\hbar$: a fixed arbitrary vertex of $\mathcal{M}_0$.



$\hbar$ has λ-edges

For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
Flipping each of the 2^k colors λ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.
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Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (1/2)

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Q. If t and s do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$, when can $t\,;s$?

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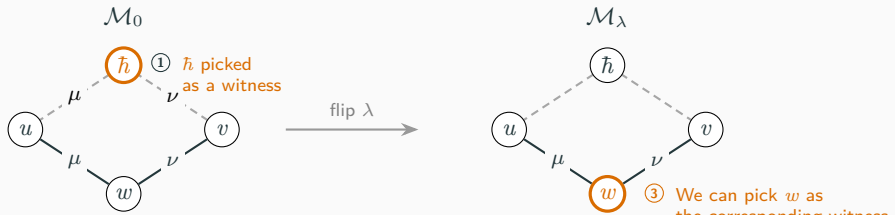
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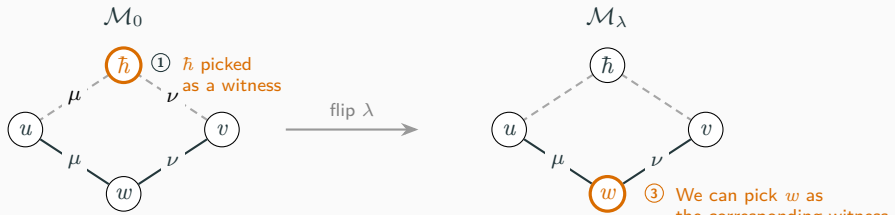
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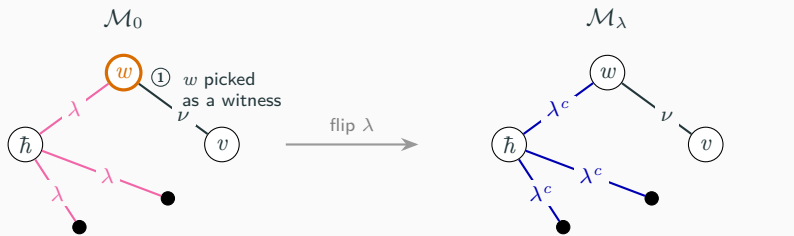
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Outline

1. Our results

2. Technical overview

3. Conclusion

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Thm. (This paper)

There are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Our proof also applies to FO^3 vs. FO^2, matrix query languages, and CoR with TC.

Key Idea the randomly colored structure $\mathcal{M}_0$ with its 2^k flipped structures.
Each composition can newly distinguish ≤ 2 of them.

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- *a superpolynomial gap over a* ***fixed*** *finite signature (e.g., one binary relation)?*
- *the domain size from which the blowup begins?*
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Thank you for your attention!

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